

The Good Morning Bed

An IoT-Based Wake-Up and Sleeping System

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PROJECT REPORT

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Abstract

The “Good Morning Bed” is a smart system built to improve how you sleep and wake up. Most regular alarm clocks wake you up too suddenly. This shock can leave you feeling groggy, tired, and stressed for the rest of the day. Our project solves this problem by creating a smart bedroom that knows when you are actually in bed.

The system uses a small computer called an ESP32, a weight sensor (FSR) to detect your body, a sound module for audio, a small motor to open window blinds, and an LED to represent the room’s air conditioning. All these parts are controlled by a custom “Awakening Algorithm.” Instead of blasting a loud alarm, this system gently wakes you up by slowly turning off the AC, opening the blinds for natural light, and playing quiet sounds. This step-by-step process helps your body wake up naturally.

1 | Introduction

1.1 Context

Getting good sleep and waking up gently are vital for your mood and brain power. Today, most people rely on loud, jarring alarms. These alarms interrupt your natural sleep cycle. Waking up too fast shocks your system. It raises stress hormones and makes you feel tired right away. We need a smarter setup that knows when you are resting and helps your body wake up smoothly.

1.2 Goals and Aims

The main goal of the Good Morning Bed is to build a highly comfortable, automated sleeping environment. Our specific goals are:

- **Improving Sleep Habits:** Replacing loud alarms with gentle, step-by-step changes in the room (like soothing sounds and natural light) to help you wake up refreshed.
- **Automating Room Climate:** Using a Green LED to act as an air conditioning unit. It will automatically turn on or off based on whether you are in bed.
- **Providing a Smooth Wake-Up:** Creating the “Awakening Algorithm.” This is a set of rules that only triggers the wake-up routine if it physically detects you are in bed.

2 | Hardware Description

This project works by combining a small computer, a weight sensor, and parts that can physically change the room (like a speaker and a motor). The diagram below shows how they all connect, followed by a breakdown of each part.

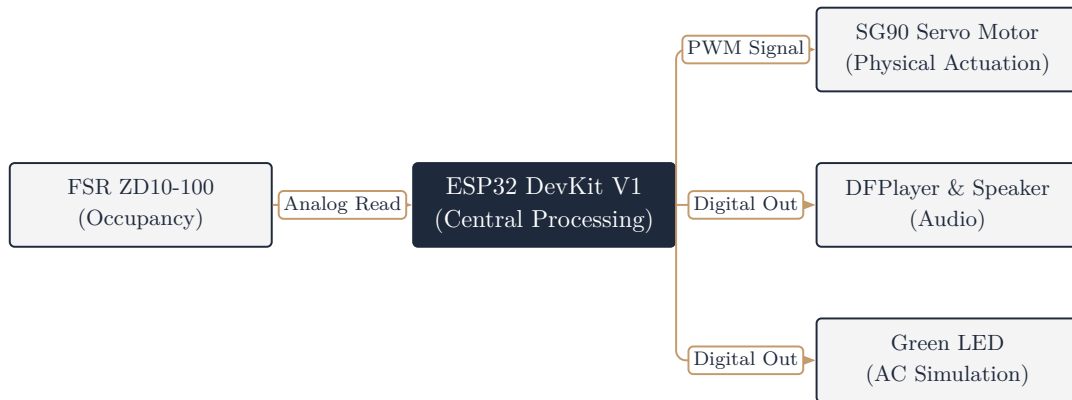


Figure 1: Simplified System Architecture Block Diagram showing expanded routing and data flow.

2.1 Microcontroller: ESP32 DevKit V1

The ESP32 is the “brain” of the Good Morning Bed. It constantly checks the sensors to see if anyone is lying down. It uses the rules in our Awakening Algorithm to decide what happens next. Based on those rules, it sends commands to play audio or turn the motor. Because the ESP32 has built-in Wi-Fi and Bluetooth, it can be easily upgraded later to connect with other smart home devices.

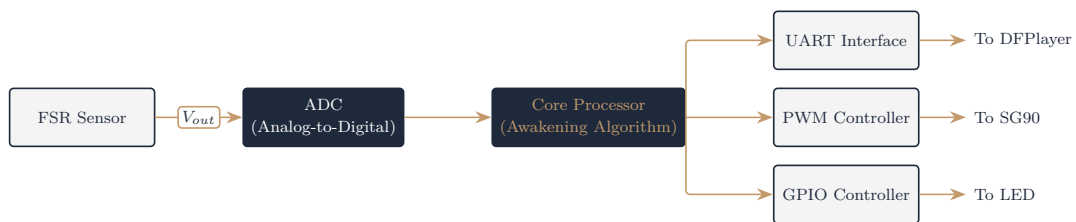


Figure 2: ESP32 DevKit V1 Internal Subsystems mapped as a clean linear signal flow.

2.2 Occupancy Sensor: ZD10-100 Thin Film Pressure Sensor

To figure out if someone is in bed, the system uses a ZD10-100 Thin Film Pressure Sensor. This is also called a Force Sensing Resistor (FSR).

How It Works: An FSR changes its electrical resistance depending on how much weight pushes down on it. When the bed is empty, its resistance is very high, acting almost like a broken wire so no power flows. When you lie down, the pressure pushes the sensor layers together. This causes the resistance to drop, allowing a voltage signal to flow. The ESP32 reads this changing voltage. We programmed a specific cut-off point so the bed knows the difference between a heavy book and an actual person.

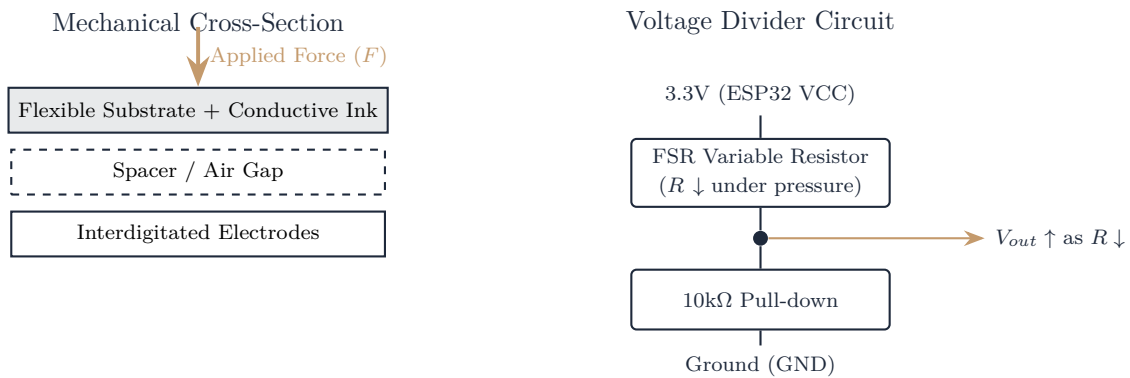


Figure 3: FSR ZD10-100 Mechanics and Voltage Divider. Downward force physically compresses the sensor's conductive layers together.

2.3 Audio Module: DFPlayer Mini and Speaker

The DFPlayer Mini is a small, cheap MP3 player. It connects directly to the ESP32 and a speaker to handle all the sound for the project. Instead of loud, annoying beeps, we loaded it with relaxing audio files. It plays white noise to help you fall asleep, and uses gentle sounds (like birds or ocean waves) that slowly get louder to help you wake up.

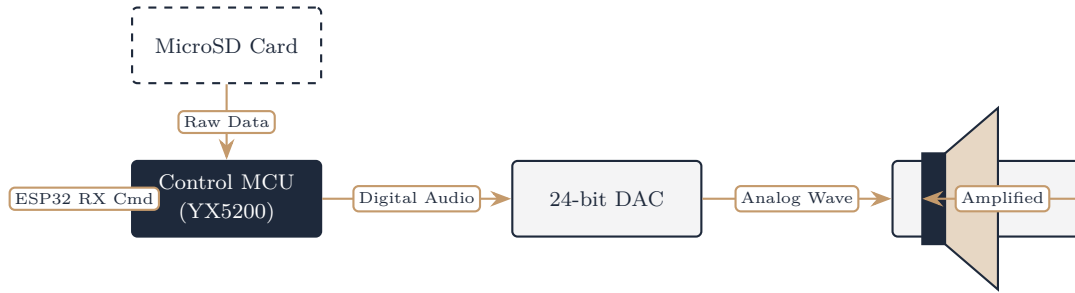


Figure 4: DFPlayer Mini Internal Data Flow representing a simplified linear conversion process.

2.4 Physical Actuator: SG90 Servo Motor

The SG90 is a small, lightweight motor that can turn halfway around (180 degrees). We use it to physically interact with the bedroom. In our setup, the motor acts like a mechanical switch that opens smart window blinds. Letting natural sunlight into the room is the best way to tell your brain to stop making sleep hormones so you wake up feeling energetic.

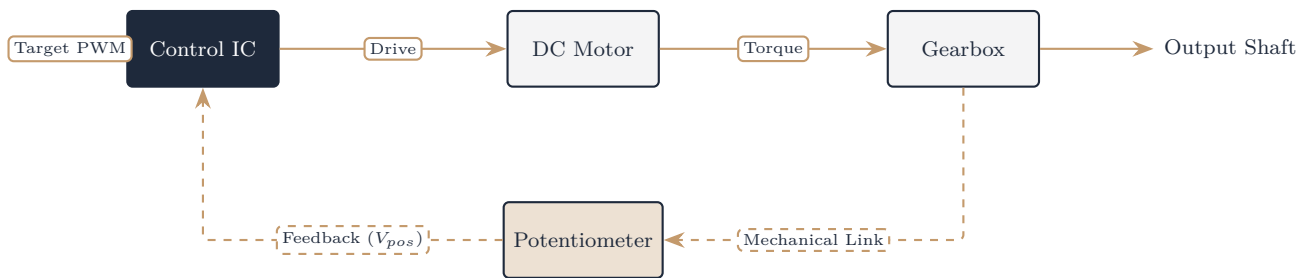


Figure 5: SG90 Servo Motor Closed-Loop Architecture utilizing clean, orthogonal feedback routing.

2.5 Visual/Environmental Actuator: Green LED

A simple Green LED acts as a visual stand-in for an Air Conditioning (AC) unit. It gives us an easy way to see what the system is doing without needing a real, heavy AC unit for testing. When the LED is ON, it means the “AC is ON” to keep the room cool for sleeping. During the wake-up routine, the LED turns OFF. This simulates the room slowly warming up, which naturally signals your body to wake up.

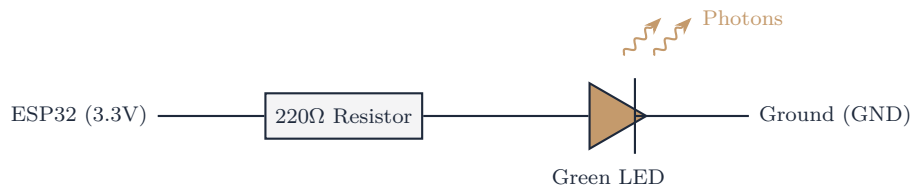


Figure 6: Green LED AC Simulation Schematic with generous spacing.

3 | System Architecture & Methodology

3.1 The Logical Flow

The system runs in a continuous loop managed by the ESP32. The logic is split into two main modes: Sleep Mode and Wake Mode. The system uses real-time data from the pressure sensor to instantly switch between them.

When you get into bed, the pressure sensor's resistance drops. The ESP32 sees this, confirms you are heavy enough to be a real person, and marks the bed as "Occupied." If it is bedtime, the ESP32 turns on the Green LED (cooling the room down) and starts playing relaxing sleep sounds.

3.2 The Awakening Algorithm

The core feature of this project is the Awakening Algorithm. This is a carefully timed routine that kicks in right before your scheduled alarm time. It ensures you wake up slowly and prevents alarms from going off if the bed is already empty.

2. Occupancy Verification: First, the ESP32 checks the pressure sensor. If the bed is empty, it completely stops the alarm. This saves power and keeps the house quiet. If you are in bed, the wake-up routine begins.
2. Environmental Shift (Climate): The ESP32 turns off the Green LED, simulating the AC turning off. The room warms up slightly, telling your body it is time to wake up.
2. Physical Actuation (Light): The ESP32 sends a signal to the servo motor, turning it 90 degrees. This simulates opening the window blinds. Natural sunlight hits your face and stops your body from making sleep hormones.
2. Auditory Escalation (Sound): Finally, the ESP32 turns on the MP3 player. It starts playing nature sounds at a very low volume. Over several minutes, the volume slowly

increases. The sounds play continuously until you physically get out of bed, at which point the system goes silent.

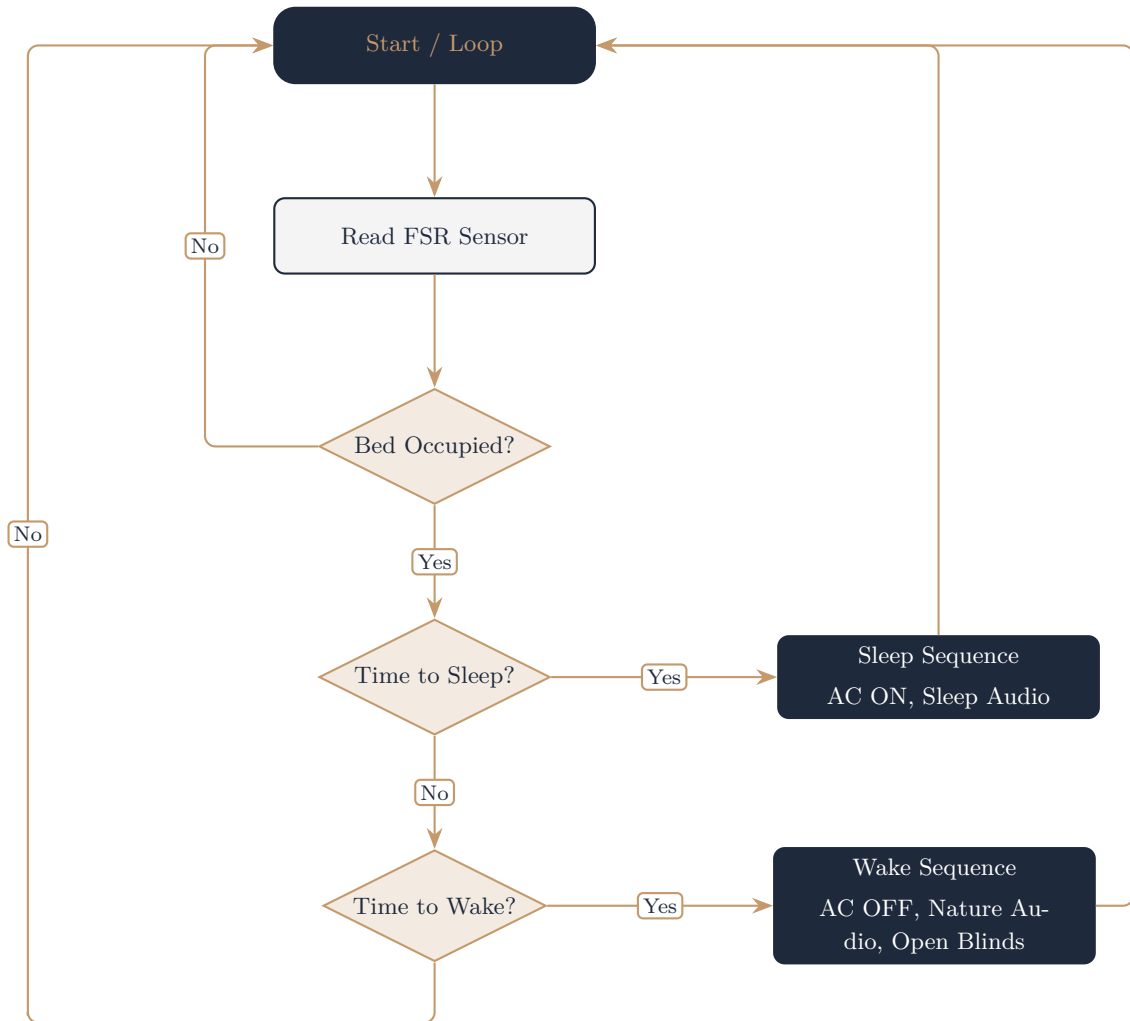


Figure 7: Integrated Algorithm Flowchart. Expanded node distances and strict orthogonal paths prevent visual clutter.

3.3 System State Machine

The entire behavior of the smart bed is controlled by a “finite state machine.” This is a programming map that shows exactly how the system moves from one state to another based on time and user actions. The flowchart below shows the rules for moving between the five main states.



Figure 8: Simplified State Machine. All transitions use embedded labels on orthogonal lines for maximum readability.

4 | Conclusion

The Good Morning Bed shows how adding simple smart devices to your home can make daily life much better. By moving away from loud, sudden alarms and using a smart “Awakening Algorithm,” the system works alongside your natural biology to help you wake up feeling refreshed. Combining a pressure sensor, a small motor, an MP3 player, and an LED creates a highly effective prototype for a modern smart bedroom. In the future, this project could easily be upgraded with machine learning. This would allow the bed to actively track your sleep cycles and wake you up during your lightest phase of sleep, making the morning routine even better.